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Underwater device for marine parameters measurement

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Leonardo Russo, Martino Saraceno, Nazareno Pinardi,

Advisor:

Prof. Paolo Schito

Co-advisors:

Mattia Belloni

Francesco Mazzeo

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Abstract: This article presents the design, construction and testing of a modular, low-cost marine sensing platform for real-time monitoring of seawater, useful in the marine research field. The system integrates pH, dissolved oxygen, oxidation-reduction potential, temperature and pressure sensors, synchronized with GPS-based positioning.

A key challenge in the development was the waterproof housing of the electronics. An initial fully 3D-printed solution proved insufficient in terms of tolerance control for the radial O-ring sealing. This issue was addressed by redesigning the enclosure sealing strategy and by introducing additional surface gasket interfaces. Field tests demonstrated the successful protection of the electronics, enabling sensor operation at depth.

The results highlight the feasibility of low-cost, open-hardware approaches for marine monitoring. Future developments will focus on adopting a dual-structure housing, composed of a water-sealed aluminum inner chamber and an external 3D-printed hydrodynamic fairing, together with external interfaces for data download and battery charging to minimize housing opening and improve long-term reliability and ease of use.

Key-words:

1. Introduction

SmartBoa is a modular and compact measurement platform designed to perform real-time monitoring of key marine parameters. Its lightweight construction, modular architecture, and user-friendly operation make it suitable for establishing a comprehensive and adaptive marine health monitoring network across the Mediterranean Sea.

The system aims to transform conventional stationary monitoring into a dynamic and distributed approach, enabling smart buoys to be installed on various types of vessels—both research and recreational—operating in critical marine zones. By continuously collecting and transmitting real-time data, the project seeks to improve the understanding of marine conditions, detect environmental changes, and support both scientific research and conservation initiatives. Furthermore, the initiative envisions citizen participation, allowing individuals to contribute to marine data collection and become active “Mediterranean Sea monitors.”

1.1. State of the Art

Current approaches to marine environmental monitoring are largely outdated and remain confined to specialized research applications. Existing systems are generally bulky, heavy, and challenging to deploy or operate, limiting their accessibility for broader use. In addition, the few available lightweight solutions are often prohibitively expensive and, in some cases, incapable of performing measurements while in motion. These limitations motivated the development of *SmartBoa*.

1.2. Advantages of SmartBoa

SmartBoa introduces a compact, cost-effective, and modular platform capable of dynamically measuring key marine parameters. The system is designed for ease of installation, operation, and storage.

The aim is to provide, unlike conventional solutions, an integrated real-time data transmission from vessels to a central database, enabling continuous monitoring and near-instantaneous visualization through interactive color maps on a user-friendly online platform. This approach not only enhances data accessibility and analysis but also fosters collaboration between scientists and citizen participants, contributing to an inclusive and scalable marine environmental observation network aimed at improving awareness and conservation of marine ecosystems.

1.3. Progetto M.A.R.E.

The *SmartBoa* concept was awarded within the framework of the *Progetto M.A.R.E.*, a marine health monitoring initiative aimed at promoting awareness of oceanic conditions by equipping researchers with dedicated instrumentation and research vessels.

Initial deployments took place among the Greek islands—an ideal testing area selected for its ecological relevance, maritime activity, and diverse environmental characteristics. As data collection and analysis progress, the project aims to expand participation to volunteer vessels operating throughout the Mediterranean, progressively extending the spatial coverage and effectiveness of the monitoring network.



Figure 1: Route of M.A.R.E. Project.

2. Design and Materials

The primary requirements for the probe were modularity, low production cost, and waterproofing. The final design fulfills all these criteria through a multi-technique manufacturing approach, combining FDM 3D printing, lamination methods, specialized resins, and advanced polymer materials.

2.1. Sonda e sistema di tenuta

The system of *SmartBoa* is composed of different components because the aim of the concept is to have a system that can be opened and modular. In particular, the main parts of the system are as follows:

- Front body;
- Rear body;
- Internal case for electronic;
- Sealing Ring;

The *Front Body* store all the sensors and a docking system is present.

The *Rear Body* has the aim of keeping the entire system stable thanks to the presence of two horizontal wings and a vertical stabilizer.

The *Internal case for electronic* contains the electronic system of *SmartBoa* .It was necessary to design it in order to provide an additional waterproof layer and protect the electronics.

The *Sealing Ring* has just to couple the front and rear parts and to ensure leakage of water within the system.



Figure 2: *SmartBoa*.

2.1.1 Modular design

One of the key points of the design was to build a modular system, to have the possibility to change the sensors depending on the measurements needed.

For this reason, the body was divided in two parts, connected by the sealing ring and 8 bolts. This design permits the system to be opened to check in case of leakages or to withdraw the SD card with the data.

Moreover, the front section was designed to allow complete disassembly and reassembly of the sensors, either to replace them for data acquisition purposes or simply to ensure easier and safer transport.

Show how the modular design works and how the things fit in, fare esploso non sarebbe male

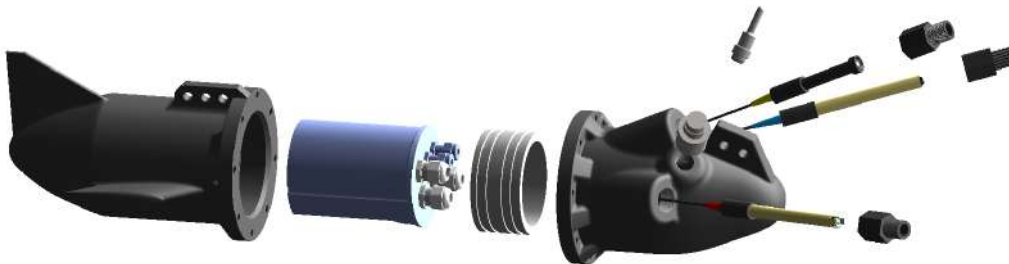


Figure 3: *Smartboa* modules explosion.

2.1.2 Sealing ring

The sealing consists of a Radial Oring fitted on a *sealing ring* a ring, that can be inserted in the system, made of gaskets grooves. In the first iteration of the design, the sealing ring was integrated directly into the front body. After tests at high pressure, problems arose due to deformation of the internal ring.

For this reason, the final design of *SmartBoa* is with a thicker sealing ring in the middle of the body, able to connect the two parts of the system and to prevent leakage inside.

Two different sealing rings were built: one made of plastic material, through 3D Printing FDM, and the other made of low cost aluminum, to reduce as much as possible the deformation.

Four radial seals are present on the part, two on the front and two on the rear. The choice of placing two seals, is not only related to a redundancy of seals, but also because it create a sort of labyrinth for the water that try to enter.

The radial seals and all the dimensional dimensions were decided thanks to *Trellerborg* online calculator.

Problems in the production phase of the aluminum sealing ring made us exit from the ISO standard, leading to one of the cause of the malfunctioning of the gaskets.

specifiche guarnizioni

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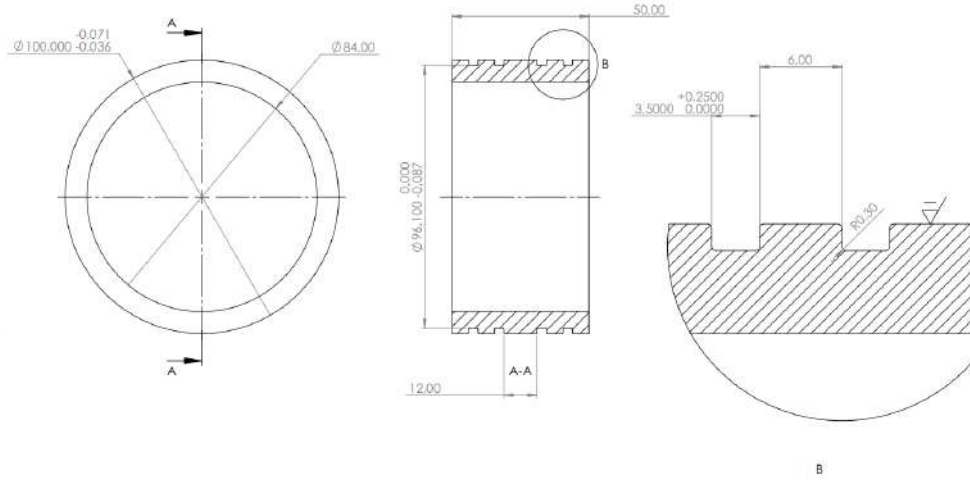


Figure 4: Sealing ring with groove highlight.

2.1.3 Smart Boa structure

The shape and structure of *SmartBoa* are designed not only to withstand the stresses at the operating conditions, but also to minimize drag.

Exploiting the *Mariotte's equation* is possible to know the state of stress of the system under the water, assuming it as a cylinder.

MARIOTTE CALCULATION

The stresses that are calculated are relatively low, so a body made of Plastic was possible. In the first part of the design a body made of metal was take into account, to reduce at minimum the thickness. It was discarded due to the high cost of production and, related to the the complex shape and material unavailability.

In addition, a layer of resin and glass fiber was placed on the external surface, this treatment increased the mechanical properties of the system, reducing also the roughness and making the body waterproof.

2.2. Cable dimensioning

SmartBoa is designed to be connected to a movement boat, for this reason a wire that connect the boat with the device is needed.

In the design phase, numerical and analytical methods were used to estimate the forces acting on the system. An analytical estimation of the drag force was carried out by initially modeling the body as a bullet and then adding the contributions to the drag coefficient (C_d) from the elements that deviate from the ideal bullet shape. These calculations were validated through a CFD analysis, which confirmed both the C_d value and the resulting drag and lift forces acting on the system.

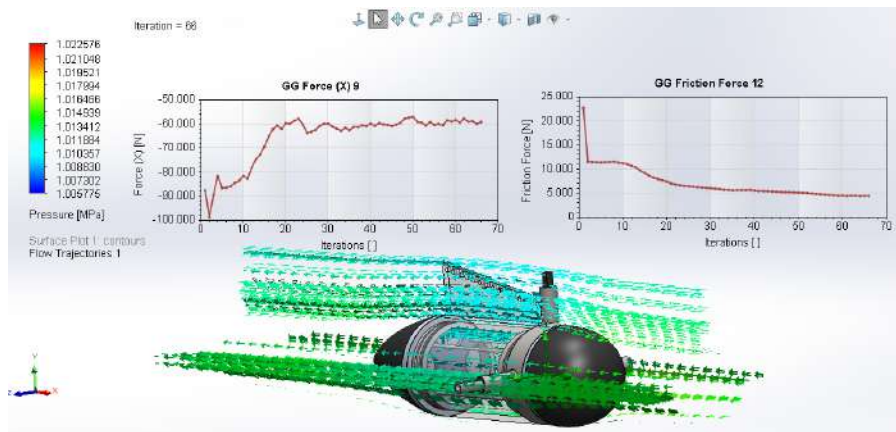


Figure 5: CFD flow and load estimation.

Once the forces were determined, it was possible to choose the correct diameter and length of the wire. For this purpose, a MATLAB script was developed which, as output, provides the angle, between the boat and the device, and the force that the wire must withstand. The model takes into account also the drag and lift forces acting



Figure 6: Angles of cable while in motion.

The length of the wire and the angle are very important because without them *SmartBoa* couldn't reach the required depth. A tuna fishing wire was chosen due to the low diameter, leading to a lower drag of the wire and the high resistance to axial load.

Redundancy is a key aspect, due to the high value of component present on the device, for this reason a second wire was connected to *SmartBoa*, on the upper surface, so in case of failure of the first wire was possible to bring back the system.

To connect the first wire, a tuna-fishing reel was used, as its drag system ensured the ability to withstand high tensile loads. For the second cable, instead, a custom reel was designed and manufactured through 3D printing. This solution provided a reliable and lightweight alternative while also serving as a backup mechanism.

2.3. Dynamic set-up and masses

One of the key aspects to consider in the design is the position of the system's center of gravity (COG) and center of buoyancy (COB), which are essential for analyzing its underwater dynamics. Accurate determination of these parameters allows for a reliable prediction of the system's stability and ensures its correct operation during submerged conditions.

2.3.1 Setup

The driving connection was placed in the front to ensure a more stable and fixed position while moving. The idea of placing it in that position derived looking at the fishing bait.

The aim of the fishing bait are to be horizontal during service and the connection between the wire and the device is placed in the front.

The hydrodynamic behavior of the wire also had to be considered. To address this, we referred to studies and practices from coastal trolling fishing (Traina Costiera). Based on these references and on the analysis of different rig configurations, additional mass was added along the wire. This allowed us to improve its stability and maintain a more controlled and efficient attitude during motion.

2.3.2 COG verification and management

The COG was determined from the CAD model, which was accurately completed with all materials and components of the system.

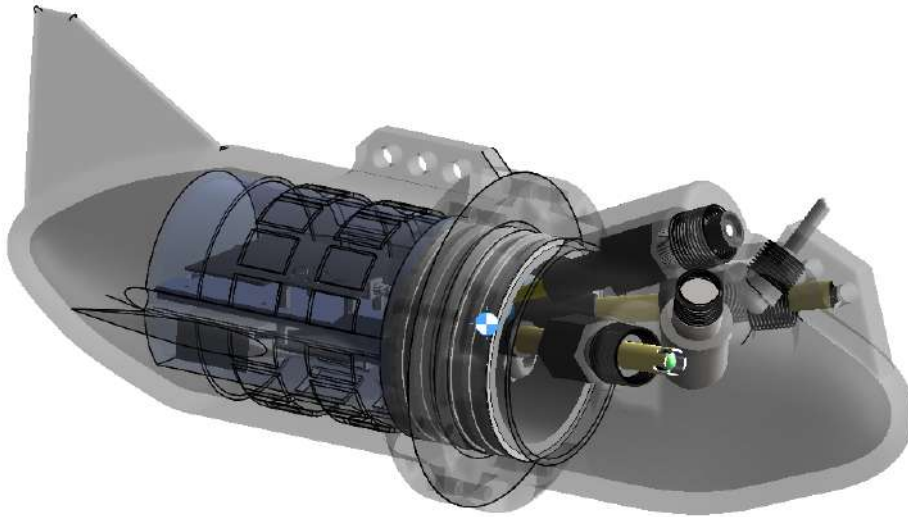


Figure 7: Graphical representation of COG.

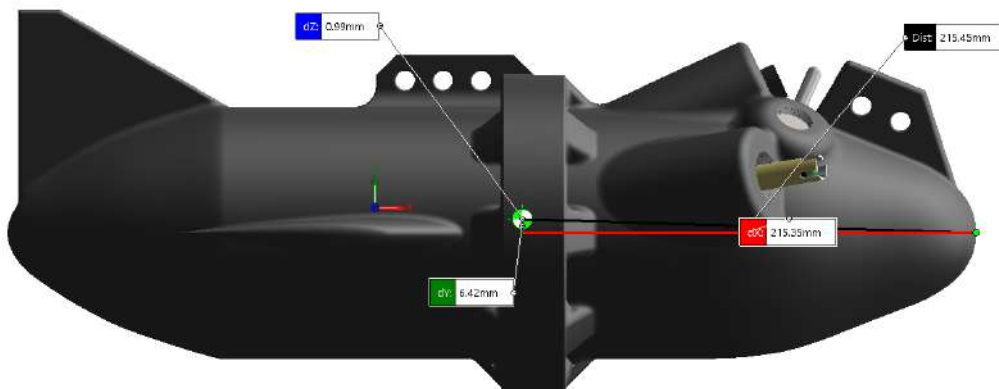


Figure 8: Coordinates respect to csys on the front.

The accurate placement of electrical and structural components leads to a minor offset of the center of gravity (COG) from the main longitudinal axis, with the COG positioned approximately at mid-length along the x-direction.

	x	y	z
Coordinates [mm]	215.35	6.42	0.99

Table 1: Coordinates respect to csys on the front.

Based on the COG analysis, balancing weights were added to further center the COG along the z-axis. These masses were also necessary to increase the overall weight of the system, ensuring proper submersion during operation.

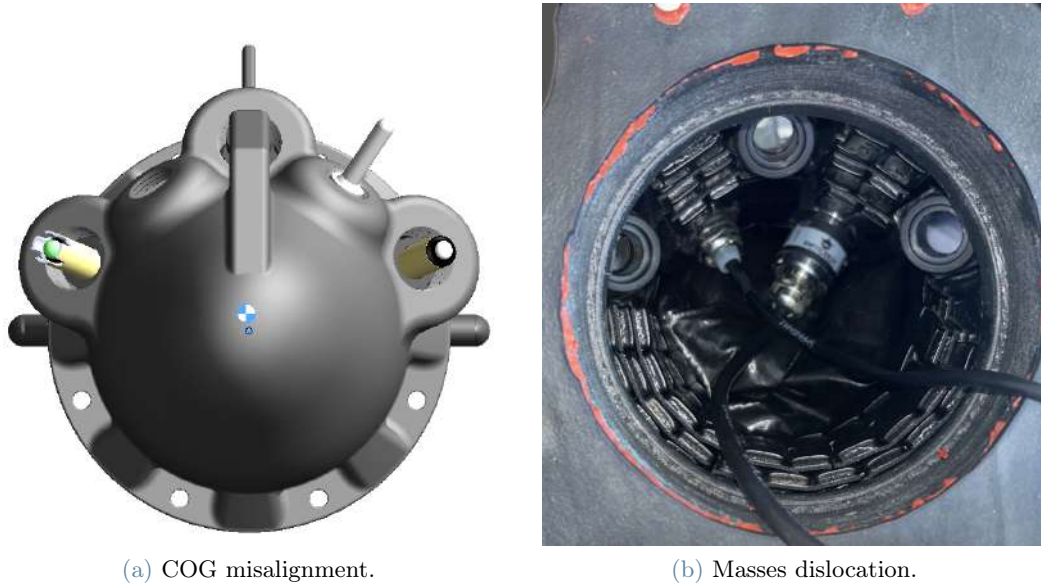


Figure 9: Mass balancing.

Owing to their ease of application and high density, wheel-balancing weights were employed for this purpose.

2.3.3 COB validation

2.4. Material Choice

The selection of materials played a critical role in meeting the design requirements of waterproofing, mechanical resistance, manufacturability, and overall cost efficiency. Different parts of the system were therefore produced using different materials and fabrication techniques, each chosen to optimize performance in its specific functional context.

The main external body was fabricated using FDM 3D printing with PLA+ and PETG polymers. These materials were selected due to their low cost, ease of manufacturing, and suitable mechanical behavior under the estimated stress conditions. Although these thermoplastic materials alone provide sufficient structural integrity, their water absorption and surface porosity required additional treatment. For this reason, a resin and fiberglass lamination layer was applied to the exterior surface, significantly improving water resistance, stiffness, and surface smoothness, thereby reducing hydrodynamic drag.

The sealing ring was produced in two variants: one in reinforced polymer through 3D printing, and a second, more rigid version in low-cost aluminum. The metallic version was introduced to minimize deformation under pressure and ensure long-term sealing performance. The radial O-rings used in the sealing interface were selected based on standard marine-grade elastomers, providing elasticity, chemical resistance, and long-term durability in submerged environments.

The wire used to connect the probe to the vessel was chosen among commercial fishing lines used for tuna fishing, due to their high axial resistance combined with a very small diameter, reducing drag during motion. The secondary safety wire, together with its custom lightweight reel manufactured via 3D printing, ensures redundancy in case of primary cable failure.

Overall, the combination of additive manufacturing, composite reinforcement, and targeted material selection enabled the construction of a modular, lightweight, and cost-efficient system capable of operating reliably in submerged and dynamic conditions.

2.5. Production techniques

The prototype was fabricated using multiple production techniques. The body and internal enclosure were produced via FDM 3D printing, with particular attention to achieving a waterproof structure. As the 3D printing process inherently leaves small voids between filament layers, two additional treatments were applied: immersion in a penetrating resin bath and lamination of glass fiber layers onto the body, using a semi-liquid resin between sheets. A turning process was employed to manufacture the sealing ring.

2.5.1 3D Printing

The main body, comprising the front and rear caps as well as the internal electronic enclosure, was fabricated using FDM 3D printing. Printing parameters were optimized in collaboration with the *Physis PEB* laboratory, with particular attention to volumetric flow rate, nozzle speed, build plate temperature, and ventilation. The structure was further optimized to enhance both mechanical strength and impermeability. Specifically, the wall loop count was set to three to increase filament density along the external surfaces, while the internal volume was filled with a Hilbert curve infill pattern to balance structural rigidity with material efficiency. Owing to the complex geometry of the prototype, tree supports were employed during printing; however, their removal caused damage to the threaded interfaces, compromising thread quality.

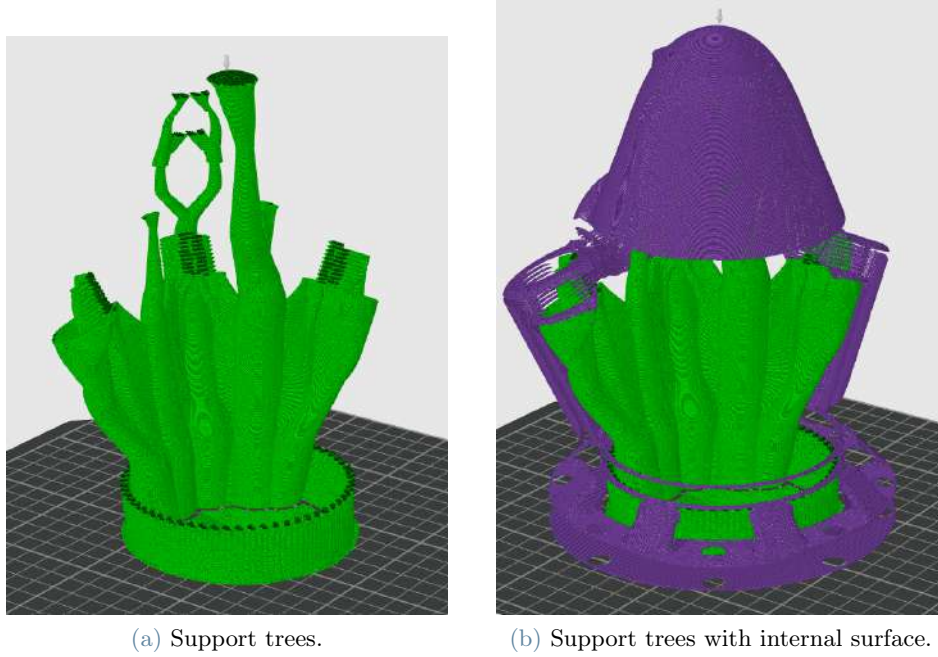


Figure 10: Trees support adopted.

The prototype was fabricated using two *X1 Carbon* 3D printers: one supplied by the Department of Mechanical Engineering at Politecnico di Milano, and the other operated by the *Physis PEB* student team of the same institution.

2.5.2 Liquid Resin and Lamination

The FDM leaves void between filament, that let water pass through, this phenomenon occurs at ambient pressure and is greater at higher one.

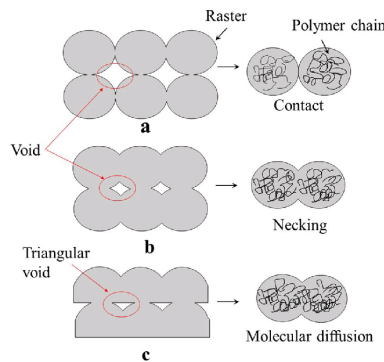


Figure 11: Three common voids in FDM parts

this phenomenon could be mitigated with the adoption of finer nozzles, higher extrusion temperature and the adoption of specific waterproof materials such as PETG.

To guarantee sealing at the thread interface while tightening the adapter, the semiliquid resin has been applied on the thread in order to eliminate void between body and adapter thread. Later on the external surface of the adapter a resin film has been applied.

3. Data Acquisition System

The goal of the system is to acquire measurements at a frequency of 1 Hz and to associate the sampled values with their geographical locations. However, since radio waves do not propagate through water, it is not possible to establish a wireless connection between the submerged device and the boat, where the position is tracked. For this reason, the current time (hours, minutes, and seconds) was recorded for each sample, in order to match the acquired data with the geographical position of the boat, which was monitored onboard. To accomplish this, it was necessary to include both a real-time clock and a micro SD card module in the circuit, to provide accurate timestamps and enable data storage.

3.1. Sensors

The monitored parameters are pH, ORP, dissolved oxygen, temperature, and pressure. Since the device is a prototype, the sensors were selected to be compatible with Arduino boards in order to build a simple and cost-effective circuit. Moreover, the sensors needed to withstand a certain level of pressure, as the device was designed to operate underwater at some depth, and to have a fast response time, since the measurements are taken as the device moves.

For this reason, the selected sensors were:

- Atlas Scientific Lab Grade pH Probe;
- Atlas Scientific Lab Grade ORP Probe;
- Atlas Scientific Lab Grade Dissolved Oxygen Probe;
- Adafruit DS18B20 Digital Temperature Sensor;
- DFRobot Gravity Analog Pressure Sensor.
- MPU6050 accelerometer

Another factor influencing the choice was the need for a waterproof interface at the points where the sensors exit the housing. To address this, Atlas Scientific Probe Pipe Fittings and Thermowells were used, as they provide an effective waterproof seal, while the pressure sensor already comes with a threaded body.

3.2. Circuit

The circuit, designed to fit within the limited space available inside the device, is based on an Arduino Nano board. While the pressure and temperature sensors can be directly connected to the Arduino, the Atlas Scientific sensors require an intermediate interface (Surveyor Boards) to output an analog signal to the controller.

The circuit was organized as follows:

- **Analog Ports:** Four analog ports were used for the pressure, pH, ORP, and dissolved oxygen sensors. The analog pressure sensor was directly connected to the board, whereas the pH, ORP, and dissolved oxygen sensors were connected through the Surveyor Boards. In addition, the DS3231 clock was connected to ports A4 and A5, using the I²C protocol.
- **Digital Ports:** One digital port was used for the DS18B20 temperature sensor, which communicates with the Arduino via the One-Wire protocol. Three additional ports (D11, D12, and D13) were used by the micro SD card module, which operates through the SPI protocol.

The circuit was powered by a 7.4 V, 2000 mAh Li-Po battery, providing at least 12 hours of continuous operation without recharging. Connections between components were soldered.

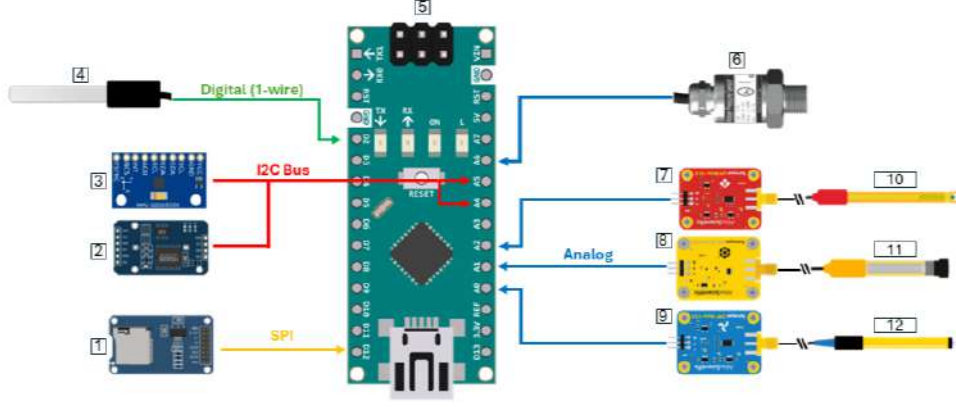


Figure 12: Circuit schematic. Components: 1) microSD storage module; 2) RTC (Real-Time Clock); 3) MPU6050 accelerometer (I²C); 4) DS18B20 temperature sensor (1-Wire); 5) Arduino Nano; 6) DFRobot pressure sensor (analog); 7) Atlas Scientific pH module with pH probe (10); 8) Atlas Scientific DO module with DO probe (11); 9) Atlas Scientific ORP module with ORP probe (12).

3.3. Circuit's Housing

To secure the circuit within the limited available space ([insert dimensions]), an openable, axially symmetric housing was designed. The circuit was mounted on both sides of the housing's internal surface, as shown in the following images:

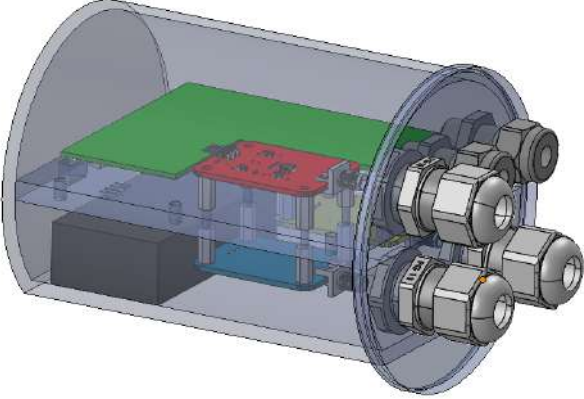


Figure 13: Housing CAD



Figure 14: Housing photo

For safety reasons, in case some water could bypass the device's waterproof sealing, the housing was fully closed. The cable interface was managed using IP68-rated cable glands, through which the cables from the five sensors were routed.

3.4. Data Analysis

As previously mentioned, the raw data acquired from the device consist of an array of coordinates, registered on board $\underline{x} = [x_1, x_2, \dots, x_m]$, and an array of samples $\underline{s} = [s_1, s_2, \dots, s_n]$, recorded by the device. The subscripts n and m are distinct, as the coordinate and sample acquisition systems operated independently; therefore, the corresponding time grids differ as well.

Indeed, the array of coordinates $\underline{x}$ is interpolated onto the samples' time grid $[t_1, t_2, \dots, t_m]$. As a result, a one-to-one correspondence between coordinates, position, and sampling time is obtained, and it's possible to generate a map.

To realize the map, Python with folium package was used to obtain an interactive map with all the parameters.



Figure 15: Map obtained in Zakynthos, near Keri

4. Testing and Sampled Data

The first field deployment of the system was carried out thanks to the *Progetto M.A.R.E.* initiative, during a route from Kerkira to Zakynthos. Unfortunately, most of the operational week on board was dedicated to troubleshooting and resolving manufacturing-related issues, and only one complete deployment could be conducted on the final day in Zakynthos.



Figure 16: System ready to be deployed.

4.1. Waterproofing

Prior to the deployment, several static tests were performed. Initially, the device was submerged without the electronic components in order to verify the watertight integrity of the housing at depth. These tests highlighted sealing issues.

The main problem was located at the waterproof interface. Although the O-ring seal was correctly dimensioned in the CAD design, the tolerances resulting from the 3D printing process were insufficient to ensure proper compression of the O-ring, ultimately preventing the housing from being fully watertight.

A similar issue was observed in the pipe fittings used to accommodate the Atlas Scientific sensors, which act as cable glands for rigid bodies. Although an NPT thread was adopted (the U.S. standard for tapered pipe threads, where sealing is achieved through the thread geometry), the dimensional tolerances introduced by the 3D-printing process were insufficient to ensure a proper thread profile. In addition, suitable threaded inserts were not available on the market.

As a consequence, the fittings could not be fully tightened into the threaded interface, allowing water to pass through the thread.

The issue was resolved by replacing the O-ring with a flat gasket, designed with perimeter holes to accommodate the housing screws. The flat gasket was bonded in place on both the front and rear interfaces of the device using a silicone-based liquid gasket sealant.

The same sealant was also applied to the pipe fitting threads to ensure proper sealing and prevent water ingress.



Figure 17: Housing CAD

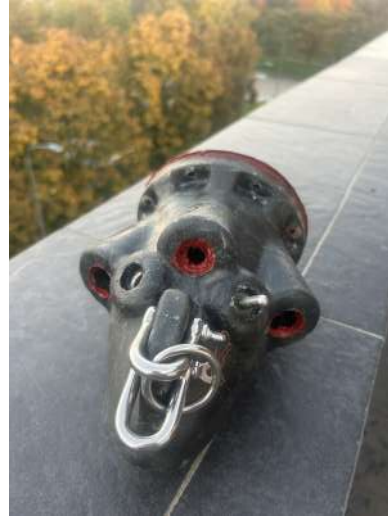


Figure 18: Housing photo

After this modification, the device achieved complete watertightness and successfully passed the static pressure test. This allowed for the first full deployment with the electronic components installed.

4.2. Total resistant force

First, the device was ballasted for 0.5 Kg over its bouyancy force. Then a first test was executed to determine the forces on the cable connecting the device to the buoy, using a luggage scale. The forces are reported in function of velocity:

- 3 knots: 14.7 N
- 4 knots: 21.5 N
- 5 knots: 39.2 N
- 6 knots: 58.9 N

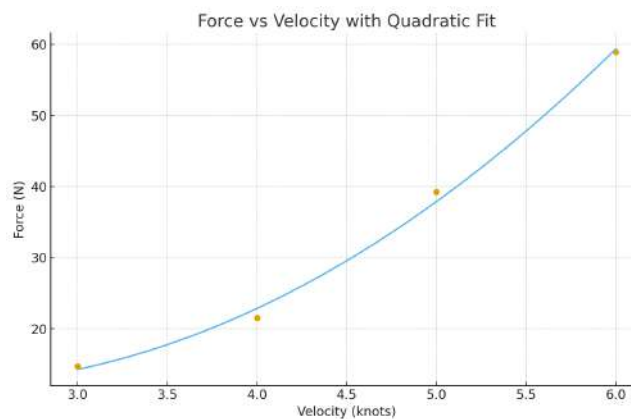


Figure 19: Velocity-Force graph

Within the 3–6 N range, the force–velocity trend appears nearly linear to the eye; however, a quadratic model provides a markedly better fit. These results confirm the predictions obtained by modeling the device as a cylindrical body, and validate the drag model in which the hydrodynamic force is proportional to the square of the velocity ($F \propto v^2$).

4.3. Sampled data

The succesful test was performed in Zakynthos, near to Keri caves, where the device was deployed for around 20 minutes, while the GPS was registering the position on board. The device stayed for all the test "al pelo dell'acqua", due to the limited time available.

The data were then matched, and the maps were computed. The trend of the sampled parameters is reported in the graph below:

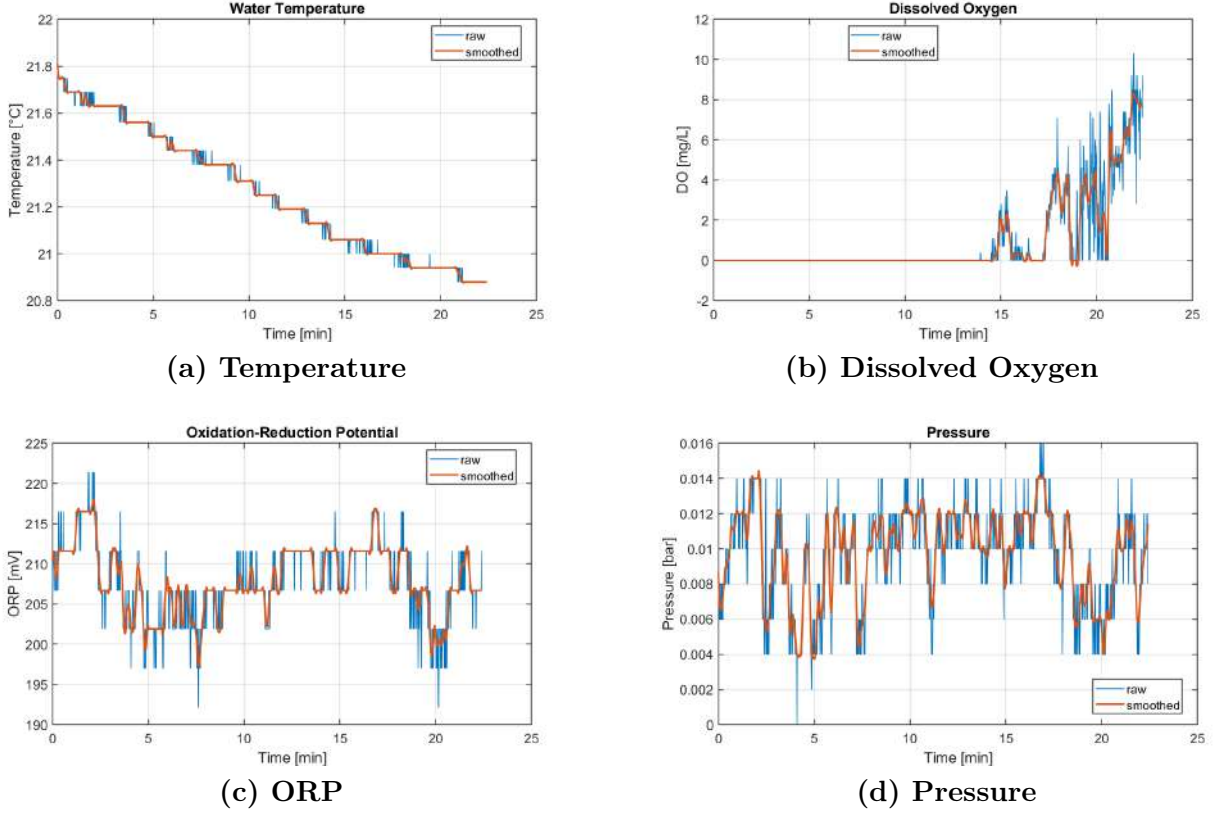


Figure 20: Time series of sensor measurements after filtering: (a) water temperature, (b) dissolved oxygen concentration, (c) oxidation–reduction potential (ORP), and (d) hydrostatic pressure.

The pH plot is not shown because the sensor malfunctioned during the deployment due to a cable fault.

The dissolved oxygen readings remain near zero for the initial 15 minutes. This is attributed to the stabilization period required by the probe after immersion, during which the internal membrane and electrolyte reach equilibrium with the surrounding water. After this transient phase, the sensor output stabilizes and provides consistent measurements.

The hydrostatic pressure values are close to zero at the surface, as expected, since the device was initially deployed at the air–water interface before submersion.

5. Conclusion

The project still has significant development margins. The FDM manufacturing process does not provide the level of dimensional tolerance required for a reliable radial O-ring seal.

Pipe fittings from Atlas scientific are not a reliable solution to prevent the leakage, the design should shift on the insulation of the cable only through the adoption of specified waterproof cable connectors.

A revision of the housing opening is planned. Since internal access should be required only rarely, future versions will integrate external ports for data download and battery charging, minimizing seal fatigue and improving operational robustness.

From these considerations, a new design should be adopted, consisting of an internal water-sealed aluminum cylinder and an external 3D-printed protective body to achieve the desired shape and hydrodynamic profile in water.

6. Bibliography and citations

Your thesis must contain a suitable Bibliography which lists all the sources consulted on developing the work. The list of references is placed at the end of the manuscript after the chapter containing the conclusions. It is suggested to use the BibTeX package and save the bibliographic references in the file `bibliography.bib`. This is indeed a database containing all the information about the references. To cite in your manuscript, use the `\cite{}` command as follows:

Here is how you cite bibliography entries: [2], or multiple ones at once: [3, 4].

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A. Appendix A

If you need to include an appendix to support the research in your thesis, you can place it at the end of the manuscript. An appendix contains supplementary material (figures, tables, data, codes, mathematical proofs, surveys, ...) which supplement the main results contained in the previous sections.